



IT6390E SEMANTIC WEB

LexNet

A Linked Open Data approach to trilingual vocabulary learning:
English, Vietnamese and Chinese

GROUP 19

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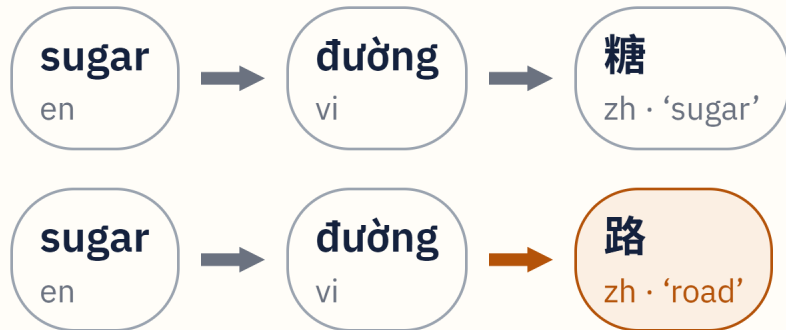
Hanoi University of Science and Technology
School of Information and Communication Technology · October 2026

THE PROBLEM

Linking languages through words goes wrong

Through a Vietnamese word

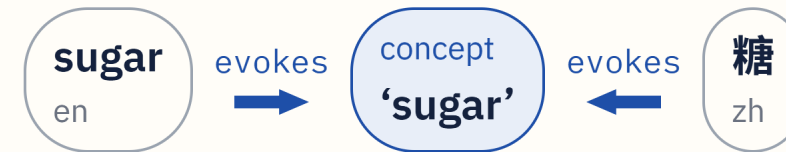
Each arrow is a dictionary translation.



Wrong match: đường also means 'road'.

Through a shared concept

Each arrow is the property `ontolex:evokes`, from a word to its concept.



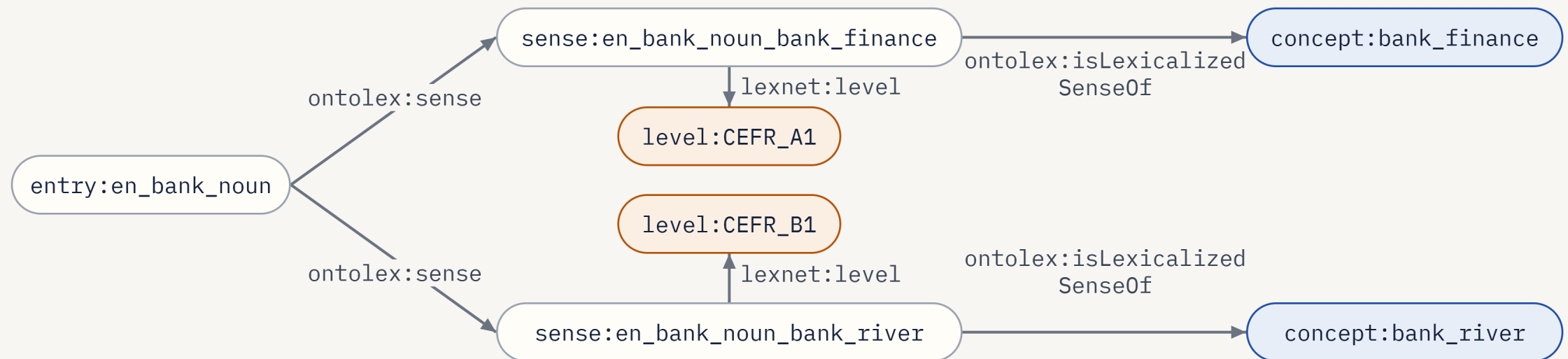
đường has one sense for 'sugar' and one for 'road'.
路 belongs only to 'road', so *sugar* reaches only 糖.

LexNet links each word sense to a shared concept, published as Linked Open Data.

STEP 1 · ONTOLOGY

Words reach concepts through their senses

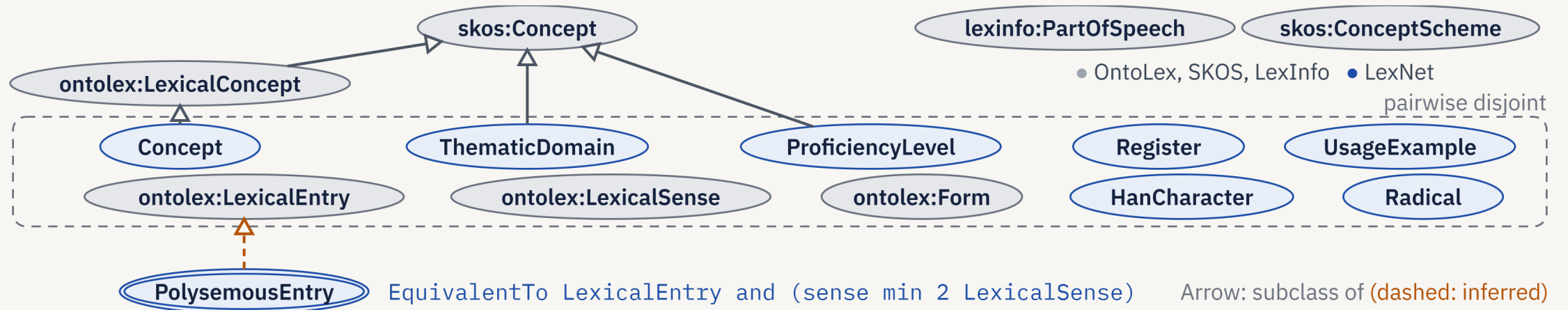
CQ3 of 9 Which meanings does a word have, and what are the level and a usage example of each?



An RDF graph of LexNet individuals: each node is an IRI and each arrow is one triple, from subject to object. `rdf:type` triples are left out.

STEP 1 · ONTOLOGY

Classes form a hierarchy, properties link instances



Property	Domain	Range	Property	Domain	Range
ontolex:sense	LexicalEntry	LexicalSense	level	LexicalSense	ProficiencyLevel
ontolex:isLexicalizedSenseOf	LexicalSense	LexicalConcept	register	LexicalSense	Register
ontolex:canonicalForm	LexicalEntry	Form	hasExample	LexicalSense	UsageExample
confusableWith	LexicalEntry	LexicalEntry	hasCharacter	Form	HanCharacter
inDomain	Concept	ThematicDomain	hasRadical	HanCharacter	Radical

Restrictions define a class and constrain the data

A defined class

```
PolysemousEntry EquivalentTo  
  LexicalEntry and  
  (sense min 2 LexicalSense)
```

Necessary and sufficient: every entry with two or more senses belongs to it, with no tagging.

The reasoner can count two senses because each has exactly one concept and all concepts are different.

Constraints

```
LexicalEntry SubClassOf  
  canonicalForm exactly 1  
LexicalSense SubClassOf  
  isLexicalizedSenseOf exactly 1  
DisjointClasses: the ten node classes  
Functional: level, register, hasRadical  
DifferentIndividuals: levels, registers,  
  concepts, radicals
```

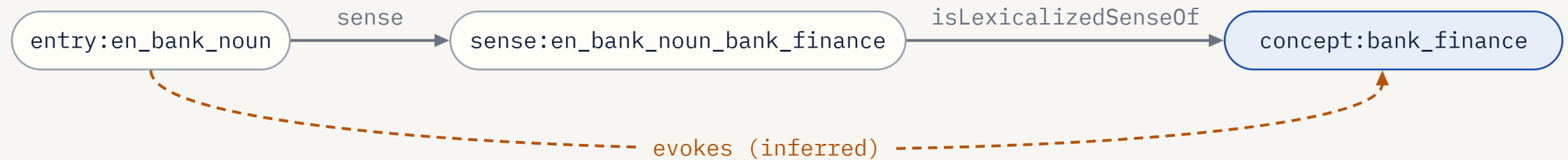
A sense with two levels, or a link to the wrong kind of node, is a contradiction.

In OWL 2 DL, but not in the EL, QL or RL profiles: reasoning needs a full DL reasoner, HermiT.

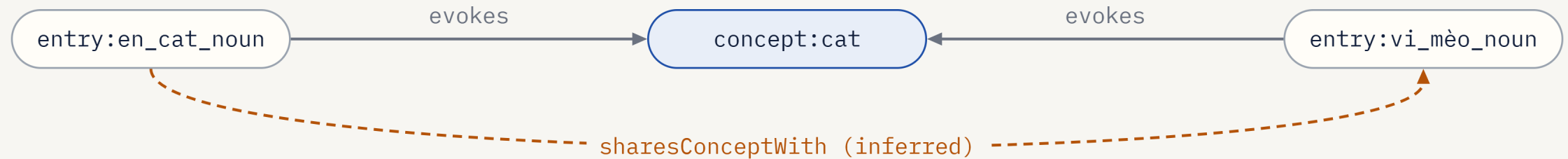
STEP 1 · ONTOLOGY

Two property chains link words through concepts

`ontolex:evokes SubPropertyChain: ontolex:sense o ontolex:isLexicalizedSenseOf`



`sharesConceptWith SubPropertyChain: ontolex:evokes o inverse (ontolex:evokes)`



Each node is an individual. Grey arrows are triples in the data, and the reasoner infers each orange dashed arrow from them. In all, this gives 1,714 links between languages and 69 synonym pairs. OWL 2 DL allows no cardinality restriction on a chained property, so PolysemousEntry counts senses.

STEP 1 · ONTOLOGY

The reasoner answers questions the data never states

Question	Definition used	What the reasoner infers
Which words are polysemous? CQ3	<code>PolysemousEntry EquivalentTo LexicalEntry and (sense min 2 LexicalSense)</code>	15 words, such as bank and đường. The class itself falls under LexicalEntry.
Which words share a concept with cat? CQ1, CQ2	<code>sharesConceptWith SubPropertyChain: evokes o inverse (evokes)</code>	mèo and 猫. In all, 1,714 links across languages and 69 synonym pairs.
Which words can be confused with lend? CQ6	<code>similarMeaning SubPropertyOf confusableWith; both Symmetric</code>	borrow. All 20 confusable pairs and 16 look-alike character pairs, both ways.
What is a tail part of? CQ4	<code>partMeronym InverseOf partHolonym</code>	cat and dog: 39 part-of links. Symmetric antonym adds 17 reverse opposites.
What is the Wikidata lexeme L3354? CQ9	<code>entry:en_bank_noun owl:sameAs wd:L3354</code>	A LexicalEntry and a PolysemousEntry, the word bank. 361 lexemes in all.

Each inference was checked with HermiT. None of these triples is in the RDF file.

Two word lists supply the candidates and levels

236

concepts in 11 everyday topics

759

words (en 237, vi 256, zh 266)

774

word senses, one
concept each

286

Chinese characters,
116 radicals

Sources

Oxford 3000 and 5000: CEFR levels

HSK 3.0 syllabus: HSK levels, pinyin

CMU Pronouncing Dictionary: IPA

Unihan, Make Me a Hanzi: characters

Process

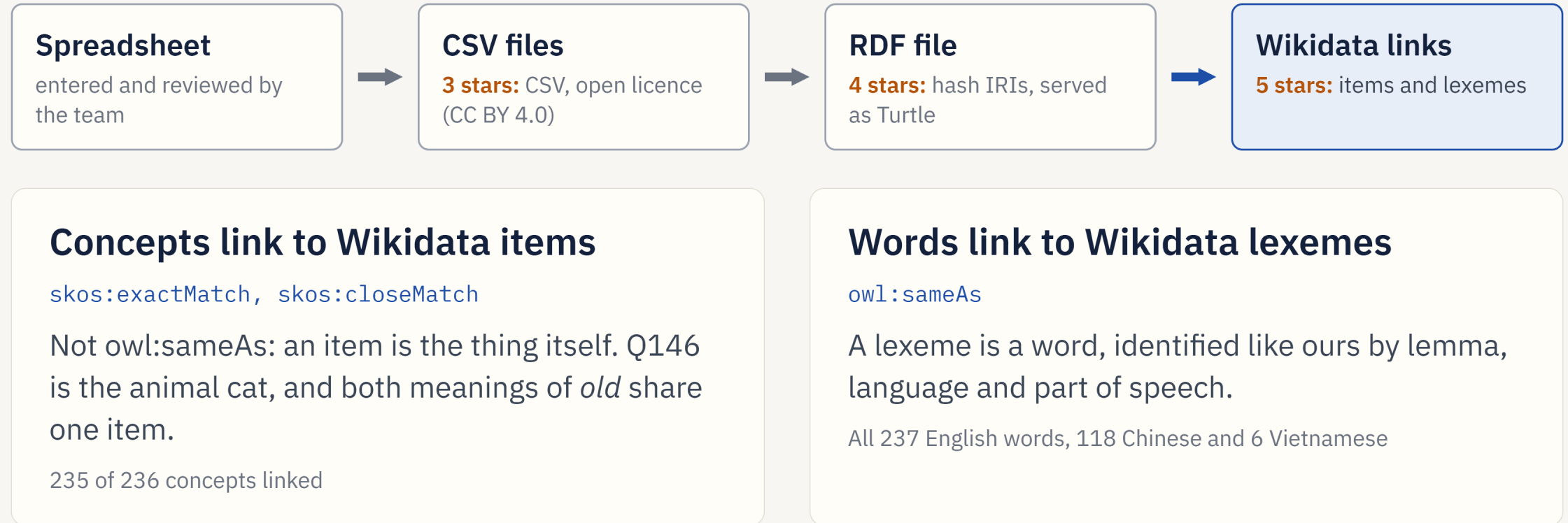
Candidates from the two word lists

Kept if all three languages have a word

Values pre-filled, every row reviewed

Usage examples written by the team

Open RDF and Wikidata links earn five stars



STEP 5 · SPARQL

A SPARQL endpoint answers the questions

```
PREFIX ontolex: <http://www.w3.org/ns/lemon/ontolex#>
SELECT DISTINCT ?word WHERE {
  ?e1 ontolex:canonicalForm/ontolex:writtenRep "猫"@zh .
  ?e1 ontolex:evokes/^ontolex:evokes ?e2 .
  FILTER(?e1 != ?e2)
  ?e2 ontolex:canonicalForm/ontolex:writtenRep ?word .
}
```

The path goes forward along evokes from 猫, then backward to **cat** and **mèò**. Arrows show the triples:



Fuseki, read-only

Loads the RDF file; the queries answer all nine competency questions.

Queries do the reasoning

Fuseki does not reason: a property path follows the chain, COUNT finds polysemous words.

Federated with Wikidata

SERVICE adds labels in seven languages, IPA, recordings and pinyin.

EVALUATION

The axioms work, and radicals only hint at meaning

HermiT finds the ontology and the data consistent, and the tests find no unwanted inference.

15

polysemous words inferred

20 + 16

confusable word pairs and look-alike
character pairs, in both directions

12 / 12

injected faults detected, one at a time

犳

Dog radical: all 5 words name mammals (狗 dog, 猫 cat, 猪 pig, 狮子 lion, 猴子 monkey)

水

Water: 6 of 9 words concern liquids; 水果 fruit, 汽车 car and 温度 temperature do not

CONCLUSION

Three languages meet at shared concepts

MODEL

Shared concepts instead of word-to-word links: *sugar* reaches only 糖.

PUBLISHING

Five-star Linked Open Data: open licence, dereferenceable IRIs, links to Wikidata.

CHECKING AND QUERYING

OWL 2 DL axioms and HermiT check the data; SPARQL answers all nine questions.

Limits: everyday vocabulary only; fewer Wikidata lexemes and recordings for Vietnamese and Chinese; the endpoint and the web application run locally.

Thank you. Questions?

github.com/Nov17th/lexnet